



COMPETENCY STANDARD

FOR

**PRODUCTION PLANNING & SUPPLY CHAIN
MANAGEMENT**

(RMG SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Production Planning & Supply Chain Management** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as a framework for providing industry-aligned training, ensuring that individuals meeting the prescribed standards through assessments are qualified to secure relevant employment opportunities.

This document was originally developed under the **Skills for Employment Investment Program (SEIP)** and later reviewed and updated with input from occupation-specific industry experts to be utilized in training initiatives under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. The document is owned by the Finance Division, Ministry of Finance, People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-PPSCM-01-G	Apply occupational health and safety (OHS) practice in the workplace	1 Identify OHS policies and procedures. 2 Apply personal health and safety practices. 3 Report hazards and risks. 4 Respond to emergencies	10
SICIP-RMG-PPSCM-02-G	Manage personal and professional development	1 Interpret personal development skills. 2 Set and meet self-development priorities. 3 Maintain professional growth and development.	10
SICIP-RMG-PPSCM-03-G	Support innovation and manage change	1 Identify needs for innovation in the workplace 2 Apply creative approach and solution 3 Support flexible and innovative ways of working 4 Adapt to emerging technological changes and opportunities	10
Total Hour			30 Hrs.

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-PPSCM-01-S	Apply Quality Procedures	1. Identify quality procedures 2. Follow quality procedure 3. Maintain standard procedure	10
SICIP-RMG-PPSCM-02-S	Apply green practices in RMG sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hour			20 Hrs.

Occupation Specific Competencies (190 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-RMG-PPSCM-01-O	Understand Production Process and Garments Manufacturing	1. Identify production and production process 2. Comprehend garments type & manufacturing methods 3. Identify major machinery used in manufacturing 4. Understand the quality control process	20
SICIP-RMG-PPSCM-02-O	Apply Knowledge of Production Planning and Supply Chain Management	1. Define production planning 2. Understand supply chain management 3. Identify supply chain process 4. Describe the relation between production planning and supply chain in RMG sector	20
SICIP-RMG-PPSCM-03-O	Perform Supply Chain Management Process	1. Identify drivers of supply chain 2. Manage inventory in supply chain 3. Follow sourcing activities 4. Describe distribution and transportation modes 5. Monitor and optimize supply chain performance 6. Ensure compliance with supply chain regulations and standards	50
SICIP-RMG-PPSCM-04-O	Apply Production Planning Tools and Techniques	1. Identify planning tools and techniques 2. Analyze production demand and forecasting 3. Identify production planning software 4. Develop and implement production schedules	50

Code	Unit of Competency	Elements of Competency	Duration (hours)
		5. Monitor and manage production processes	
SICIP-RMG- PPSCM-05-O	Manage Supply Chain Uncertainty	1. Identify sources of uncertainty in the supply chain 2. Assess the impact of supply chain disruptions 3. Develop strategies to mitigate risks and uncertainties 4. Implement contingency plans 5. Communicate with stakeholders	30
SICIP-RMG- PPSCM-06-O	Perform Real Life Case Study	1. Identify a relevant real-life case study 2. Analyze the case study's background and context 3. Evaluate solutions and outcomes 4. Develop actionable recommendations	20
Total Hours			190 Hrs.

COMPETENCY STANDARD: PRODUCTION PLANNING & SUPPLY CHAIN MANAGEMENT

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-PPSCM-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 2.2 Common health issues are recognized. 2.3 Common safety issues are identified.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 <u>Emergency response plans and procedures</u> are responded to. 4.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Emergency response plans and procedures	2.1 Firefighting procedures 2.2 Earthquake response procedures

	2.3 Emergency response plans and procedures 2.4 Medical and first aid
3. First aid procedure	3.1 Washing of open wound 3.2 Washing chemically infected area 3.3 Applying bandage 3.4 Taking appropriate medicine
4. Personal protective equipment	4.1 Safety glasses 4.2 Ear plugs 4.3 Gloves 4.4 Apron 4.5 Helmet 4.6 Mask 4.7 Safety shoes

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource implications	The following resources must be provided: 3.1 Workplace (simulated or actual) 3.2 Personal protective equipment (PPE) 3.3 Firefighting equipment 3.4 Emergency response manual 3.5 First aid kits

	3.6 Stationery 3.7 Learning manual.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 applied personal health and safety practices 1.3 reported hazards and risks 1.4 responded to emergencies.
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MANAGE PERSONAL AND PROFESSIONAL DEVELOPMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-PPSCM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to manage personal and professional development. It specifically includes the tasks of interpreting personal development skills, setting and meeting self-development priorities and maintaining professional growth and development.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret personal development skills	1.1 Objectives of personal development skills are described. 1.2 <u>Personal development skills</u> are identified. 1.3 Intra and Interpersonal relationships are maintained in the course of managing oneself. 1.4 Self-analysis is performed and personal development needs are identified.
2. Set and meet self-development priorities	2.1 Tasks are prioritized to achieve personal, team and organizational goals and objectives. 2.2 <u>Resources</u> are utilized efficiently and effectively to manage work priorities and commitments. 2.3 Economic usage and maintenance of facilities are followed as per established procedures.
3. Maintain professional growth and development	3.1 Pro activeness/zeal is demonstrated in fulfilling personal and professional growth requirements. 3.2 <u>Trainings and career opportunities</u> are identified and accessed based on job requirements. 3.3 <u>Recognitions</u> are sought/ received and demonstrated as proof of career advancement. 3.4 <u>Licenses and/or certifications</u> relevant to the job and career are obtained and renewed

Range of variables:

Variables	Range (may include but not limited to)
1. Personal Development skills	1.1 Problem-solving 1.2 Self-confidence 1.3 Adaptability 1.4 Integrity 1.5 Work ethic 1.6 Pro-activeness

2. Resources	2.1 Human 2.2 Financial 2.3 Technology
3. Trainings and career opportunities	3.1 Participation in training programs 3.1.1 Technical 3.1.2 Supervisory 3.1.3 Managerial 3.1.4 Continuing Education 3.2 Serving as Resource Persons in conferences and workshops
4. Recognitions	4.1 Recommendations 4.2 Citations 4.3 Certificate of Appreciations 4.4 Commendations 4.5 Awards 4.6 Tangible and Intangible Rewards
5. Licenses and/or certifications	5.1 National Certificates 5.2 Certificate of Competency 5.3 Support Level Licenses 5.4 Professional Licenses

Curricular Content Guide

1. Underpinning knowledge	1.1 Importance of personal development skills. 1.2 Organizational policies relevant to training and professional growth. 1.3 Company operations, procedures and standards. 1.4 Resources in work environment.
2. Underpinning skill	2.1 Utilizing and improving personal development skills 2.2 Maintaining Intra and Interpersonal relationship 2.3 Utilizing communication skills 2.4 Prioritizing tasks in accordance with work commitment. 2.5 Utilizing resources efficiently and effectively
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure

	4.4 Workplace documents, signs and symbols
	4.5 Codes of conduct
	4.6 Projector
	4.7 Stationery
	4.8 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 attained personal development skills. 1.2 maintained intra and interpersonal relationship in the course of managing oneself. 1.3 prioritized tasks according to work commitments. 1.4 identified training and career opportunities. 1.5 completed trainings based on the requirements of the industries. 1.6 acquired and maintained licenses and/or certifications according to the requirement of the qualification.
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: SUPPORT INNOVATION AND MANAGE CHANGE	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-PPSCM-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to support innovation and manage change. It specifically includes the tasks of identifying needs for innovation in the workplace, applying creative approach and solution, supporting flexible and innovative ways of working and adapting to emerging changes and opportunities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify needs for innovation in the workplace	1.1. The need for <u>innovation</u> in scope of work is recognized 1.2. The value of <u>innovative practices</u> in the workplace is recognized 1.3. Existing way of working in the workplace are interpreted. 1.4. Drawbacks of existing way of working are identified. 1.5. Benefits of change are identified to make it consistent with organizational requirements 1.6. Realistic timelines and targets for implementation of changes are set.
2. Apply creative approach and solution	2.1 Opportunities and creative approaches to implement innovation practices are identified 2.2 Creative approaches of coworkers pertaining to work practices are analyzed and incorporated 2.3 Innovation in accordance with organizational requirements changes are implemented
3. Support flexible and innovative ways of working	3.1 Maximum innovative opportunities are promoted. 3.2 Work assignments to facilitate innovative work skills are organized. 3.3 Team members are provided with guidance and coaching on innovation in the workplace. 3.4 Models of innovative work practice are provided and discussed 3.5 Appropriate environment for learning and innovation is maintained.
4. Adapt to emerging technological changes and opportunities	4.1 Usages of different <u>technologies</u> is determined based on job requirements. 4.2 Appropriate technology is selected as per work specifications. 4.3 Relevant technology is effectively used in carrying out functions. 4.4 Appropriate implementation tools are used as per task

	requirement.
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Range of variables:

Variables	Range (may include but not limited to)
1. Innovative practices	1.1. Self-directed support to implement new ideas 1.2. Collaborative arrangement to apply new method of working 1.3. Making scope of work more efficient to use new technology
2. Innovation	2.1. New ideas 2.2. Different ideas 2.3. New methods of doing work 2.4. Use of new Technology
3. Technologies	3.1. Office technology 3.2. Industrial technology 3.3. System technology 3.4. Information technology 3.5. Training technology

Curricular Content Guide

1. Underpinning knowledge	1.1. Definition of innovation 1.2. Current practice in own scope of work 1.3. Workplace procedures 1.4. Support required to generate creative ideas 1.5. Difference between innovation and creativity 1.6. Common effects of change and innovation in the workplace 1.7. Industrial and organizational context of change 1.8. Organization's policies, plans, procedures and structure 1.9. Knowledge of resources required by the organization's operations 1.10. Awareness on technology and its function 1.11. Operating instructions 1.12. Applicable software 1.13. Company policy in relation to relevant technology 1.14. Technology adaptability
2. Underpinning skill	2.1. Identifying resources required for creativity and innovation 2.2. Contributing to brainstorming session 2.3. Identifying issues and concerns of one's scope of work 2.4. Encouraging co-workers to generate and develop ideas 2.5. Evaluating potential obstacles to and opportunities for creativity and innovation 2.6. Sharing of best practices related to innovation and

	creativity 2.7. Applying relevant technology 2.8. Using update machine and Software 2.9. Acquiring troubleshooting skills Identifying issues 2.10. Identifying current industry standard diagnostic tools 2.11. Describing common malfunctions and resolutions. 2.12. Determining the root cause of a routine malfunction
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 5.1 Workplace (simulated or actual) 5.2 Projector 5.3 Stationery 5.4 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified need for innovation in the workplace 1.2 applied creative approach and solution 1.3 supported flexible and innovative ways of working 1.4 adapted to emerging technological changes and opportunities
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY QUALITY PROCEDURES	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-PPSCM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply quality procedures. It specifically includes the tasks of identifying quality procedures, following quality procedure and maintaining standard procedure.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify quality procedures	1.1. <u>Manuals</u> and guidelines for RMG production processes are collected as per sample specifications. 1.2. The importance of manuals in ensuring quality standards is recognized. 1.3. Instructions and procedures for quality control are identified. 1.4. Required information is collected from manuals. 1.5. Performance measurement systems are identified.
2. Follow quality procedure	2.1 Instructions and procedures are followed strictly and duties are performed in accordance with demand of <u>quality improvement system</u> . 2.2 Concept of supplying product or service to meet the <u>customer quality requirements</u> is understood and accordingly applied. 2.3 Conformance to specifications is ensured. 2.4 Defects are detected and reported to the relevant authority according to standard operating procedures
3. Maintain standard procedure	3.1 Performance is assessed at regular interval. 3.2 Specifications and standard operating procedures are established 3.3 Quality of product is checked and verified. 3.4 Quality control and quality assurance system procedures for each job are followed. 3.5 Conformance to specification is ensured in every case at all situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. Manuals	1.1 Buyers' specification manual 1.2 Compliance manual

	1.3 Maintenance procedure manual 1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Signs and symbols, instruction manuals
2. quality improvement system.	2.1. Quality inspection 2.2. Testing 2.3. Quality control. 2.4. Quality assurance 2.5. Total Quality Management
3. customer quality requirements	3.1. Performance 3.2. Features 3.3. Reliability 3.4. Conformance 3.5. Aesthetics 3.6. Durability

Curricular Content Guide

1. Underpinning Knowledge	1.1 Importance of maintaining quality 1.2 quality, quality assurance, quality control, quality inspection, quality improvement and total quality control. 1.3 Process and procedures for improving and maintaining quality 1.4 Procedures for addressing defects. 1.5 Record keeping within the quality improvement system in workplace 1.6 Factors, which affect successful implementation of the quality systems and procedures.
2. Underpinning Skills	2.1. Maintaining good quality 2.2. Eliminating poor quality 2.3. Understanding the meaning of the key terms - quality, quality assurance, quality control, quality inspection, quality improvement and total quality control. 2.4. Improving and maintaining quality 2.5. Addressing defects and procedures 2.6. Recording within the quality improvement system in workplace. 2.7. Implementing quality systems and procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace

4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Map/globe 4.3 Projector 4.4 Stationery 4.5 Learning manual.
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified quality procedures 1.2 followed quality procedure 1.3 maintained standard procedure
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN RMG SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-PPSCM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in RMG sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices in RMG (Ready-Made Garment)</u> manufacturing are explained. 1.2 Key terms and symbols in RMG production processes are identified. 1.3 <u>Sources of environmental impacts</u> during RMG manufacturing activities are identified. 1.4 RMG manufacturing activities use are explained. 1.5 <u>Ways of mitigating environmental impacts</u> in RMG manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in RMG manufacturing are documented. 2.2 <u>Recyclable and non-recyclable items</u> in RMG production processes are identified. 2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented. 2.4 Sustainable alternatives in RMG manufacturing are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste in RMG manufacturing</u> are identified. 3.2 Hazardous waste in RMG production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in RMG manufacturing processes. 1.2 6R for waste management in RMG production: 1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste generation.

	1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in RMG manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in RMG production with low environmental impact. 1.4 Recycling of materials used in the RMG manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.
2. Sources of environmental impacts	2.1 Air Pollution in RMG Manufacturing 2.2 Water Pollution in RMG Manufacturing 2.3 Soil Degradation in RMG Manufacturing 2.4 Noise Pollution in RMG Manufacturing 2.5 Waste Generation in RMG Manufacturing 2.6 Resource Depletion in RMG Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in RMG production processes. 3.2 Adopting renewable energy sources for RMG manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 Recyclable Items in RMG Manufacturing 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 Non-Recyclable Items in RMG Manufacturing 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.

5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in RMG manufacturing 1.6 Hazardous waste in RMG production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in RMG manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in RMG manufacturing 2.6 Disposing hazardous waste in RMG production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: UNDERSTAND PRODUCTION PROCESS AND GARMENTS MANUFACTURING	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG- PPSCM-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitude required to understand production process and garments manufacturing. It specifically includes the tasks of identifying production and production process, comprehending garments type & manufacturing methods, identifying major machinery used in manufacturing and understanding the quality control process.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Production and Production Process	1.1. Production is defined and understood. 1.2. Key stages of the <u>production process</u> are identified and described. 1.3. Flow of materials through the production process is understood and explained. 1.4. <u>6M of production</u> are identified and explained.
2. Comprehend Garments Type & Manufacturing Methods	2.1 Different <u>garment types</u> are identified and classified. 2.2 Basic <u>manufacturing methods</u> is understood and described. 2.3 <u>Types of fiber, yarn & fabric</u> is identified and explained.
3. Identify Major Machinery Used in Manufacturing	3.1 <u>Types of sewing machines</u> used in garment manufacturing are identified. 3.2 Stitch and seam type is identified and explained. 3.3 Role of automation in modern sewing machines is explained.
4. Understand the Quality Control Process	4.1 Quality control objectives are identified and understood. 4.2 Quality control checkpoints are identified throughout the production process. 4.3 Key stages of the quality control process are identified. 4.4 <u>Quality control tools and techniques</u> are recognized and explained. 4.5 Role of quality control in production is described.

Range of Variables

Variable	Range (Includes but not limited to):
1. Production process	1.1 Input

	1.2 Output 1.3 Transformation process 1.4 Feedback
2. 6M of production	2.1 Man 2.2 Machine 2.3 Material 2.4 Method 2.5 Money 2.6 Management
3. Garment types	3.1 Woven garments 3.2 Knit garments 3.3 Sweater
4. Manufacturing methods	4.1 Measurement Chart 4.2 Sketch/Pattern 4.3 Sample 4.4 Cutting 4.5 Print and Embroidery 4.6 Sewing 4.7 Washing 4.8 Finishing 4.9 Packing
5. Types of fiber, yarn & fabric	Fiber 5.1 Natural 5.2 Man-made Yarn 5.3 Carded yarn 5.4 Combed yarn Fabric 5.5 Woven 5.6 Knit 5.7 Non-woven
6. Types of sewing machines	6.1 Single needle lock stitch 6.2 Single needle chain stitch 6.3 Double needle lock stitch 6.4 Double needle chain stitch 6.5 Overlock 6.6 Feed of the arm 6.7 Flat lock 6.8 Multi needle Kansai 6.9 Button stitch 6.10 Button hole 6.11 Snap button
7. Quality control tools and techniques	7.1 Average Quality Level (AQL) 7.2 Pareto analysis 7.3 Fishbone diagram 7.4 Control charts 7.5 Quality audits

	7.6 Defect per Hundred Units (DHU)
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand the definition and purpose of production in manufacturing. 1.2 Explain the flow of materials through each production stage 1.3 Identify the 6Ms: Man, Machine, Material, Method, Measurement, and Mother Nature 1.4 Understand basic manufacturing methods 1.5 Identify various sewing machines and their uses. 1.6 Understand the role of automation in sewing machines. 1.7 Identify QC checkpoints at different production stages 1.8 Recognize common QC tools and techniques
2. Underpinning Skills	2.1 Manage material flow 2.2 Apply the 6M (Man, Machine, Material, Method, Measurement, Environment). 2.3 Classify garment types and apply manufacturing methods. 2.4 Utilize fibers, yarns, fabrics, and sewing machines. 2.5 Demonstrate sewing machines functions. 2.6 Apply quality control objectives, checkpoints, and tools. 2.7 Ensure quality through control processes. 2.8 Report hazards and maintain a safe environment. 2.9 Use automation and digital tools in production.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Sample garments 4.4 Sample fiber, yarn and fabric 4.5 Sewing machines

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified production and production process 1.2 comprehended garments type & manufacturing methods
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	1.3 identified major machineries used in manufacturing 1.4 understanding the quality control process
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY KNOWLEDGE OF PRODUCTION PLANNING AND SUPPLY CHAIN MANAGEMENT	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG- PPSCM-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitude required to apply knowledge of production planning and supply chain management. It specifically includes the tasks of defining production planning, understanding supply chain management, identifying supply chain process, and describing the relation between production planning and supply chain in RMG sector.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Define production planning	1.1. Production planning is understood to meet manufacturing targets in the RMG sector. 1.2. Key aspects of production planning are identified and explained. 1.3. Relationship between demand forecasts and production plans is recognized. 1.4. Production planning bottlenecks and capacity constraints are identified.
2. Understand supply chain management	2.1 Concept of supply chain is defined and understood. 2.2 <u>Key components of the supply chain</u> are identified and described. 2.3 The relationship between procurement, production, and distribution is explained. 2.4 <u>Supply chain risks</u> are explained.
3. Identify supply chain process	3.1 <u>Supply chain processes</u> are identified and understood. 3.2 Raw material procurement is recognized. 3.3 Logistics and distribution processes are recognized. 3.4 Role of information flow across the supply chain is identified. 3.5 Importance of quality control and assurance processes is understood.
4. Describe the relation between production planning and supply chain in RMG sector	4.1 Relationship between production planning and the supply chain is described. 4.2 Role of production schedules and delivery logistics is described. 4.3 <u>Supply chain disruptions</u> are explained in relation to their impact on production planning. 4.4 Interaction between production deadlines and distribution schedules is explained.

	4.5 Importance of information sharing between production and supply chain teams is described.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Key components of the supply chain	1.1 Suppliers 1.2 Manufacturers 1.3 Distributors 1.4 Retailers 1.5 Customer
2. Supply chain risks	2.1 Delays 2.2 Disruptions 2.3 Quality issues 2.4 Inventory risks 2.5 Procurement risks
3. Supply chain processes	3.1 Inventory management 3.2 Logistics 3.3 Demand forecasting 3.4 Supply chain integration (push/pull view)
4. Supply chain disruptions	4.1 Delays in raw material delivery 4.2 Logistics issues 4.3 Production capacity issues 4.4 Power and utility interruptions 4.5 Lack of manpower

Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand the purpose of production planning to achieve manufacturing targets. 1.2 Recognize how demand forecasts influence production decisions and planning. 1.3 Define and understand the concept of supply chain in the RMG context. 1.4 Explain the interrelationship between procurement, production, and distribution processes. 1.5 Understand various supply chain processes from raw material sourcing to finished goods delivery. 1.6 Understand the role of information flow in improving coordination. 1.7 Describe the relationship between production planning and supply chain operations. 1.8 Explain how supply chain disruptions impact production planning.
2. Underpinning Skills	2.1 Address key aspects of production planning. 2.2 Understand the relationship between demand forecasts and production plans.

	2.3 Recognize production bottlenecks and address capacity constraints. 2.4 Demonstrate the relationship between procurement, production, and distribution. 2.5 Understand supply chain processes and their execution. 2.6 Manage raw material procurement efficiently. 2.7 Handle logistics and distribution processes effectively. 2.8 Utilize information flow to optimize supply chain performance. 2.9 Implement quality control and assurance processes across the supply chain.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual)

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 defined production planning 1.2 understood supply chain management 1.3 identified supply chain process 1.4 described the relation between production planning and supply chain in RMG sector
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM SUPPLY CHAIN MANAGEMENT PROCESS	Nominal Duration: 50 Hrs.	Unit Code: SICIP-RMG-PPSCM-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitude required to perform supply chain management process. It specifically includes the tasks of identifying drivers of supply chain, managing inventory in supply chain, following sourcing activities, describing distribution and transportation modes, monitoring and optimizing supply chain performance, and ensuring compliance with supply chain regulations and standards.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify drivers of supply chain	1.1 <u>Supply chain drivers</u> in the RMG sector are identified and described. 1.2 Relationship between supply chain drivers is explained. 1.3 <u>Supply chain drivers' metrics</u> are defined and described. 1.4 Best practices for improving <u>supply chain performance</u> in the RMG sector are understood.
2. Manage Inventory in Supply Chain	2.1 Inventory management concepts are identified and explained. 2.2 <u>Types of inventories</u> are identified and described. 2.3 <u>Inventory control methods</u> are interpreted and understood. 2.4 <u>Inventory tracking systems</u> are identified and explained for managing stock levels. 2.5 Relationship between inventory levels and production planning is understood and described.
3. Follow Sourcing Activities	3.1 Suppliers are identified based on specific sourcing requirements. 3.2 Supplier evaluation and selection are carried out. 3.3 Supplier negotiations are conducted to agree on price, payment terms and delivery schedules. 3.4 Contract management is handled by ensuring that contracts are drafted. 3.5 Purchase orders are issued according to organizational requirements. 3.6 Supplier relationship management is maintained by fostering positive.
4. Describe Distribution and Transportation Modes	4.1 <u>Distribution networks</u> are identified. 4.2 Different <u>modes of transportation</u> are described. 4.3 Advantages and disadvantages of each transportation

	mode are explained. 4.4 Regulatory and compliance requirements for transportation and distribution are identified. 4.5 Impact of transportation on supply chain performance is understood.
5. Monitor and optimize supply chain performance	5.1 Key supply chain performance indicators are identified and monitored. 5.2 Data from supply chain operations is analyzed. 5.3 Supply chain processes are optimized. 5.4 Inventory levels are optimized by using forecasting tools. 5.5 Cost reduction strategies are implemented to optimize the supply chain. 5.6 Risk management practices are followed.
6. Ensure compliance with supply chain regulations and standards	6.1 Relevant supply chain regulations and standards are identified. 6.2 Compliance audits are conducted to verify that all supply chain partners. 6.3 Documentation requirements are understood and maintained. 6.4 Compliance issues are addressed promptly by identifying non-compliance. 6.5 Third-party audits and certifications are managed.

Range of Variables

Variable	Range (Includes but not limited to):
1. Supply chain drivers	1.1 Facility 1.2 Inventory 1.3 Transportation 1.4 Information 1.5 Sourcing 1.6 Pricing
2. Supply chain drivers' metrics	2.1 Lead time 2.2 Inventory flow 2.3 Order fulfillment 2.4 Capacity 2.5 Product verity 2.6 Production service level 2.7 Safety inventory 2.8 Forecast horizon 2.9 Profit margin
3. Supply chain performance	3.1 Including lean manufacturing 3.2 Just-in-time (JIT) inventory 3.3 Integrated systems 3.4 Supply chain responsiveness 3.5 Efficient supply chain 3.6 Achieving strategic fit

4. Types of inventories	4.1. Raw materials 4.2. Work-in-progress (WIP) 4.3. Finished goods 4.4. Machine components
5. Inventory control methods	5.1. Economic Order Quantity (EOQ) 5.2. Just-In-Time (JIT) 5.3. Lot sizing with multiple products 5.4. Safety stock
6. Inventory tracking systems	6.1. Barcode scanning 6.2. Radio Frequency Identification (RFID) 6.3. ERP systems
7. Distribution networks	7.1. Direct shipping 7.2. Indirect shipping 7.3. Hybrid 7.4. Cross docking
8. Modes of transportation	8.1. Road transport 8.2. Rail transport 8.3. Air 8.4. Sea 8.5. Pipeline 8.6. Package carrier 8.7. Hybrid model
9. Key supply chain performance indicators	9.1. Lead time 9.2. Pricing 9.3. High service level 9.4. Order accuracy 9.5. Inventory turnover 9.6. Delivery performance

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and describe major supply chain drivers 1.2 Understand performance metrics for each driver 1.3 Understand key inventory management concepts 1.4 Interpret common inventory control methods 1.5 Identify and explain inventory tracking systems 1.6 Identify suppliers based on sourcing requirements 1.7 Identify different distribution networks and their structure 1.8 Understand regulatory and compliance requirements for logistics 1.9 Identify and monitor key performance indicators (KPIs) 1.10 Identify relevant supply chain regulations and standards
2. Underpinning Skills	2.1 Classify and describe different types of inventories. 2.2 Apply inventory control methods.

	2.3 Manage inventory tracking systems to control stock levels. 2.4 Evaluate and select suppliers through a structured process. 2.5 Monitor key supply chain performance indicators. 2.6 Analyze data from supply chain operations for insights. 2.7 Optimize supply chain processes for efficiency. 2.8 Use forecasting tools to optimize inventory levels. 2.9 Implement cost reduction strategies to enhance supply chain efficiency.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified drivers of supply chain 1.2 managed inventory in supply chain 1.3 followed sourcing activities 1.4 described distribution and transportation modes 1.5 monitored and optimized supply chain performance 1.6 ensured compliance with supply chain regulations and standards
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY PRODUCTION PLANNING TOOLS AND TECHNIQUES	Nominal Duration: 50 Hrs.	Unit Code: SICIP-RMG- PPSCM-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitude required to apply production planning tools and techniques. It specifically includes the tasks of identifying planning tools and techniques, analyzing production demand and forecasting, identifying production planning software, developing and implementing production schedules, and monitoring and managing production processes.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Planning Tools and Techniques	1.1. <u>Production planning tools</u> are identified and explained. 1.2. <u>Planning techniques</u> are identified and explained in the context of ensuring smooth production flow. 1.3. ERP and MRP systems are identified and explained. 1.4. Lean manufacturing principles and Six Sigma techniques are described.
2. Analyze Production Demand and Forecasting	2.1. Production demand is analyzed by reviewing historical sales data, seasonal trends, and market forecasts. 2.2. <u>Planning tools</u> are used to support demand analysis and production scheduling. 2.3. <u>Demand forecasting techniques</u> are applied to predict future production needs. 2.4. Capacity planning is performed to ensure that <u>production resources</u> . 2.5. Market trends and customer preferences are evaluated. 2.6. Forecast accuracy is reviewed by comparing predicted and actual demand.
3. Identify Production Planning Software's	3.1. Types of <u>production planning software</u> are identified. 3.2. Features of production planning software are described. 3.3. Benefits of using production planning software are explained. 3.4. Role of production planning software in optimizing manufacturing operations is understood.
4. Develop and Implement Production Schedules	4.2 Production requirements are identified based on demand forecasts, customer orders, and inventory levels. 4.3 Scheduling tools (spreadsheets) are used to create

	and track production timelines. 4.4 Resource availability is verified and planned. 4.5 Production schedules are applied as required.
5. Monitor and Manage Production Processes	5.1 Key performance indicators (KPIs) are tracked and analyzed. 5.2 Production data from machines, sewing line and operators are collected and reviewed. 5.3 Production processes are monitored to ensure they meet quality, cost, and lead time.

Range of Variables

Variable	Range (Includes but not limited to):
1. Production planning tools	1.1. TNA plan 1.2. Gantt charts 1.3. MRP (Material Requirements Planning) 1.4. MPS (Master Production Schedule) 1.5. ERP (Enterprise Resource Planning) systems
2. Planning techniques	2.1 Work breakdown structures 2.2 Forecasting 2.3 Capacity planning 2.4 Target and efficiency
3. Planning tools	3.1 Priority rules 3.2 Gantt charts 3.3 MRP systems 3.4 ERP software
4. Demand forecasting techniques	4.1 Moving averages 4.2 Exponential smoothing 4.3 Trend analysis 4.4 Seasonal adjustments
5. Production resources	5.1. Labor 5.2. Machine and equipment 5.3. Materials
6. Production planning software	6.1. Fast react 6.2. System Application and Product (SAP) 6.3. General Sewing Data (GSD)
7. Resource	7.1. Raw materials 7.2. Labor 7.3. Equipment
8. Key performance indicators (KPIs)	8.1. Production rate 8.2. Defect rates 8.3. Downtime 8.4. Inventory turnover 8.5. Material storage and retrieve

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and explain production planning tools 1.2 Recognize planning techniques 1.3 Identify and understand ERP (Enterprise Resource Planning) and MRP (Material Requirements Planning) systems. 1.4 Analyze production demand using historical sales data, seasonal trends, and market forecasts. 1.5 Review forecast accuracy by comparing actual demand with predictions. 1.6 Identify production requirements based on forecasts, customer orders, and current inventory. 1.7 Apply production schedules to manufacturing operations according to plan. 1.8 Identify different types of production planning software 1.9 Explain benefits of using such software 1.10 Identify and track key performance indicators (KPIs)
2. Underpinning Skills	2.1 Use planning techniques to ensure smooth production flow. 2.2 Apply Lean manufacturing principles and Six Sigma techniques. 2.3 Analyze production demand by reviewing historical sales, seasonal trends, and market forecasts. 2.4 Use planning tools for demand analysis and production scheduling. 2.5 Apply demand forecasting techniques to predict future production needs. 2.6 Perform capacity planning to ensure resource availability. 2.7 Use scheduling tools (e.g., spreadsheets) to create and track production timelines. 2.8 Apply production schedules as required.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 ERP and MRP system 4.4 Planning tools 4.5 Production planning software

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified planning tools and techniques 1.2 analyzed production demand and forecasting 1.3 identified production planning software's 1.4 developed and implemented production schedules 1.5 monitored and managed production processes
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MANAGE SUPPLY CHAIN UNCERTAINTY	Nominal Duration: 30 Hrs.	Unit Code: SICIP-RMG-PPSCM-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitude required to manage supply chain uncertainty. It specifically includes the tasks of identified sources of uncertainty in supply chain, assessing the impact of supply chain disruptions, developing strategies to mitigate risks and uncertainties, implementing contingency plans and communicating with stakeholders.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Sources of Uncertainty in Supply Chain	1.1 External factors affecting the supply chain are identified. 1.2 Demand uncertainty is recognized. 1.3 <u>Supply variability</u> is understood. 1.4 Logistical challenges are identified. 1.5 <u>Technological disruptions</u> are understood. 1.6 Internal factors within the organization are identified. 1.7 Risk of stockouts or overstocking is understood.
2. Assess the impact of supply chain disruptions	2.1 Impact on customer satisfaction is evaluated. 2.2 Impact on inventory levels is understood. 2.3 Operational impact of supply chain disruptions is analyzed. 2.4 Financial risks posed by supply chain disruptions are assessed. 2.5 Reputational impact of supply chain disruptions is evaluated.
3. Develop strategies to mitigate risks and uncertainties	3.1 <u>Risk assessment processes</u> are developed. 3.2 Risk or disruption is identified. 3.3 Diversification strategies are developed to ensure safety inventory and buffer stocks. 3.4 Supplier relationships are strengthened, with measures to build strong, long-term partnerships. 3.5 Regular monitoring and updates are ensured to implement strategies.
4. Implement contingency plans	4.1 Contingency plans are created. 4.2 Alternative suppliers and routes are identified and engaged as per the contingency plan. 4.3 Inventory and resource allocation are adjusted based on the contingency plan. 4.4 Monitoring and tracking of the situation are conducted as per plan. 4.5 Regulatory and compliance requirements are maintained.

5. Communicate with stakeholders	5.1 Stakeholders are identified and categorized 5.2 Stakeholders are communicated and discussed following communication framework . 5.3 Supply chain coordination is established to avoid bullwhip effect . 5.4 Alternative solutions are proposed and communicated. 5.5 Regular updates are provided throughout the duration of the uncertainty. 5.6 Documentation of communications is maintained.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Supply variability	1.1 Supplier lead times 1.2 Quality control inconsistencies 1.3 Supplier reliability 1.4 Availability of raw materials 1.5 Geopolitical issues affecting supply sources 1.6 Changes in supplier costs or pricing strategies
2. Technological disruptions	2.1 System failures in supply chain software 2.2 Technological obsolescence of current systems 2.3 Integration issues between different software platforms
3. Risk assessment processes	3.1 Risk Identification 3.2 Risk Analysis 3.3 Risk Evaluation 3.4 Risk Mitigation Planning 3.5 Risk Monitoring and Review
4. Communication framework	4.1 Customer Relationship Management (CRM) 4.2 Internal Supply Chain Management (ISCM) 4.3 Supplier Relationship Management (SRM)
5. Bullwhip effect	5.1 Inaccurate forecasting 5.2 Excessive inventory 5.3 Increased costs

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify external factors affecting the supply chain 1.2 Recognize demand uncertainty and its causes 1.3 Understand supply variability due to lead times, supplier reliability, or raw material scarcity. 1.4 Understand the risk of stockouts and overstocking and their effects. 1.5 Understand inventory impact due to supply chain disruptions. 1.6 Identify potential risks or disruptions using data, past trends, and forecasts.
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	1.7 Identify and engage alternative suppliers and logistic routes in advance. 1.8 Identify and categorize key stakeholders.
2. Underpinning Skills	2.1 Evaluate the impact on customer satisfaction and inventory levels. 2.2 Analyze the operational impact of supply chain disruptions. 2.3 Assess the financial risks posed by disruptions. 2.4 Evaluate the reputational impact of disruptions. 2.5 Develop risk assessment processes and strategies. 2.6 Identify risks or disruptions early. 2.7 Develop diversification strategies for safety inventory and buffer stocks. 2.8 Create contingency plans and engage alternative suppliers and routes as needed.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified sources of uncertainty in supply chain 1.2 assessed the impact of supply chain disruptions 1.3 developed strategies to mitigate risks and uncertainties 1.4 implemented contingency plans 1.5 communicated with stakeholders
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.

	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.		
Unit of Competency: PERFORM REAL LIFE CASE STUDY		Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG- PPSCM-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitude required to perform real life case study. It specifically includes the tasks of identifying a relevant real-life case study, analyzing the case study's background and context, evaluating solutions and outcomes and developing actionable recommendations.			

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify a relevant real-life case study	1.1 A relevant real-life case study is identified and selected based on its alignment with the learning objectives. 1.2 <u>Key issues or challenges</u> faced in the case study are described. 1.3 Solutions or strategies implemented in the case study are explained. 1.4 Relevance of the case study to the industry or field of study is explained. 1.5 Key learnings or insights from the case study are identified and summarized.
2. Analyze the case study's background and context	2.1 Background of the case study is reviewed and summarized. 2.2 Context of the case study is analyzed. 2.3 Relevant stakeholders involved in the case study are identified. 2.4 <u>Key historical factors</u> or precedents are examined. 2.5 Assumptions and biases in the case study's background are recognized.
3. Evaluate solutions and outcomes	3.1 Supply chain drivers are identified based on the case study. 3.2 <u>Obstacles achieving strategic fit</u> is identified based on case study. 3.3 Possible solutions to the identified obstacles are assessed. 3.4 Best solution for achieving cost minimization or maximizing the value of the supply chain is determined. 3.5 Sustainability of the solution is evaluated considering long-term impacts on the supply chain.
4. Develop actionable recommendations	4.1 Supply chain networks are re-designed based on the findings from the case study if required.

	4.2 Sourcing decisions are made by evaluating suppliers, costs, lead times, and quality. 4.3 Warehousing and inventory control systems are improved, with recommendations. 4.4 Facility requirements and technological upgrades are assessed if required. 4.5 Impact on stakeholders is considered when developing recommendations. 4.6 Sustainability considerations are integrated into the recommendations if required. 4.7 Actionable steps and timelines for implementing the recommendations are provided if required.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Key issues or challenges	1.1 Financial constraints 1.2 Operational inefficiencies 1.3 Technological limitations or disruptions 1.4 Regulatory or legal issues 1.5 Internal resistance or organizational culture problems
2. Key historical factors	2.1 Political and Governmental Changes 2.2 Economic Developments 2.3 Cultural and Social Movement 2.4 Environmental and Geographic Influences
3. Obstacles achieving strategic fit	3.1 Variety of the products 3.2 Decreasing of product life cycle 3.3 Globalization 3.4 Increasing demanding customer 3.5 Changing business environment

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and select a real-life case study relevant to the learning objectives and industry context. 1.2 Describe the key issues or challenges encountered in the selected case. 1.3 Understand and explain the relevance of the case study to real-world industry practices. 1.4 Identify and summarize key insights and learnings derived from the case study. 1.5 Recognize any assumptions or biases that might affect interpretation or conclusions. 1.6 Identify specific supply chain drivers within the case. 1.7 Identify obstacles to achieving strategic fit 1.8 Assess needs for facility upgrades or technological improvements to enhance efficiency.
2. Underpinning Skills	2.1 Review and summarize the background of the case study.

	2.2 Analyze the context of the case study. 2.3 Assess possible solutions to identified obstacles. 2.4 Evaluate the sustainability of the solution considering long-term impacts. 2.5 Make sourcing decisions by evaluating suppliers, costs, lead times, and quality. 2.6 Improve warehousing and inventory control systems with recommendations. 2.7 Provide actionable steps and timelines for implementing recommendations, if necessary.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified a relevant real-life case study 1.2 analyzed the case study's background and context 1.3 evaluated solutions and outcomes 1.4 developed actionable recommendations
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Production Planning & Supply Chain Management
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Desktop Computer with Planning software	25
2.	Laptop	1
3.	Different Garment Sample (Woven & Knit)	30
4.	Different Fabric Sample (Woven & Knit)	30
5.	Display Board	2
6.	Single needle lock stitch machine	1
7.	Over lock machine	1
8.	Flat lock machine	1
9.	Multimedia projector screen	1
10.	Scanner	1
11.	Printer	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Production Planning & Supply Chain Management the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.